

Fluorescence distributions in combinatorial models of amyloid fibrils composed of split-YFP, Sup35p, and CFP

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Abstract

We derive exact finite-length fluorescence statistics for four combinatorial null models of amyloid fibrils: split-YFP alone, split-YFP with Sup35p, FRET YFP/CFP, and FRET YFP/CFP with Sup35p. In their natural units, the split-YFP statistics $R_{SY}(n) := 2E(n)/n$ and $R_{SY,s}(n) := 2E(n)/n$ converge to $2/3$ and $1/3$, while the FRET statistics $R_F(n) := E(n)/n$ and $R_{F,s}(n) := E(n)/n$ converge to $1/2$ and $2/9$. Because these ratio families represent different physical quantities, they should not be compared as fluorescence yields. For split-YFP systems, bivariate generating functions and transfer matrices give exact probability distributions $P(n, k)$ as closed binomial sums and rational functions. They also yield closed forms for $k \cdot 2^k$ -weighted shallow diagonals of Pascal's triangle related to OEIS entries A127976 and A095977. A common rational recurrence structure unifies the Sup35p models and gives exact expectations and variances; independent Markov chain arguments verify the results. For FRET systems, linearity of expectation gives a finite interaction-radius baseline, with $E_r(n)/n \rightarrow 2rp_{CPY}$ under an additive unit-weight cutoff. We also give unequal-concentration limits and Monte Carlo spot checks. The models are transparent combinatorial baselines, not mechanistic descriptions of fibril growth.

Keywords: Amyloid fibrils, combinatorial probability, generating functions, fluorescence resonance energy transfer, split-YFP, transfer matrices

1 Introduction

Amyloids are fibrillar protein aggregates characterized by their distinctive β -sheet structure, where proteins self-assemble into highly ordered stacks (Chiti and Dobson 2006). For the purposes of our mathematical analysis, we abstract these complex biochemical structures into a simplified model where individual proteins, denoted A,B,C,..., form sequential patterns in their stacking arrangements (e.g., ABBACCBCBCA...). This abstraction turns a biophysical setting into a tractable combinatorial null model rather than a mechanistic description of fibril growth.

Our combinatorial approach follows the general framework of analytic combinatorics (Flajolet and Sedgewick 2009, Ch. I). We study amyloids of different compositions: split Yellow Fluorescent Proteins (split-YFP), a technique based on bimolecular fluorescence complementation (Hu et al. 2002; Kerppola 2006); split-YFP together with *Saccharomyces cerevisiae* eukaryotic translation release factor Sup35p, a prion-forming protein (Patino et al. 1996); and YFP/Cyan Fluorescent Protein (CFP) systems under donor excitation that can produce Förster resonance energy transfer (FRET) (Förster 1948; Patterson et al. 2000). A specific quantity of interest is the expected number of *fluorescing locations*—points between two adjacent proteins in the amyloid stack at which a fluorescent interaction occurs (a fusion in the split-YFP settings, a $\{C, Y\}$ adjacency in the FRET settings). We derive the full probability distributions $P(n, k)$ for the number of fluorescing sites and closed-form expressions for the expected values using standard generating functions, transfer matrices, Markov chains, and regular expressions. The symbol $\mathcal{A}(n, k)$ is used locally within each model for the number of length- n amyloids with k fluorescing sites.

In the split-YFP settings, the adjacency blocking constraint leads to weighted shallow-diagonal sums such as $\sum_{k=1}^{\lfloor n/2 \rfloor} k 2^k \binom{n-k}{k}$, which we evaluate via generating function techniques (Wilf 2006). Direct summation and Markov chain arguments yield the same closed forms, providing mutual verification. The closed-form identities that

arise turn out to match known OEIS sequences A127976 and A095977 ([OEIS Foundation Inc.](#); [OEIS Foundation Inc.](#)); the contribution of this work is not the identities themselves, but the exact null-model statistics obtained by applying this familiar combinatorial pipeline to a family of related fluorescence models.

Several modeling assumptions should be stated at the outset. First, proteins are assumed to be independently and uniformly distributed in the amyloid sequence. Real amyloid assembly is templated and cooperative ([Chiti and Dobson 2006](#)), so this i.i.d. assumption is a baseline against which non-random assembly effects could be detected. Second, Sup35p is used here purely as an abstract inert third species: the model does not attempt to describe Sup35p’s own prion biology, only the effect of inserting a non-fluorescent stack partner. Third, in the split-YFP sections, we model fluorophore reconstitution as a local pairing rule; the rule is motivated by bimolecular fluorescence complementation being effectively irreversible on experimental timescales ([Hu et al. 2002](#); [Kerppola 2006](#)), but abstracts away geometric and kinetic details of real fibril growth. Fourth, in the FRET sections, we initially restrict interactions to nearest neighbors in the fibril, then record the expected-value extension to a finite interaction radius in Section 6. In practice, the inter-strand spacing in cross- β amyloid structure is approximately 0.47 nm ([Sunde et al. 1997](#)), while reported Förster radii for common CFP–YFP variants are on the order of 5 nm ([Patterson et al. 2000](#); [Lakowicz 2006](#)), so the Förster scale is on the order of ten amyloid spacings. The effective interaction along a fibril nevertheless depends on fluorophore geometry and orientation and decays steeply with distance, so this scale should not be read as a hard interaction radius. The nearest-neighbor restriction is therefore a deliberately simplified $r = 1$ baseline, not the quantitative model one would compare directly to experiment. Finally, the equal-concentration assumption ($p = 1/2$ or $1/3$ per protein type) is adopted throughout the main sections; limiting unequal-concentration formulas are discussed in Section 7.

Throughout, K_n denotes the fluorescence count in a length- n amyloid under the model currently being discussed, $E(n) = \mathbb{E}[K_n]$, and $P(n, k) = \mathbb{P}(K_n = k)$. The counting symbol $\mathcal{A}(n, k)$ is model-local: it denotes the number of length- n amyloids with $K_n = k$ in the current section, with the empty word convention $\mathcal{A}(0, 0) = 1$ when bivariate generating functions are used. In split-YFP models, K_n counts fused A–B pairs, so $2K_n$ proteins participate in fluorescence. In FRET models, K_n counts additive C–Y interaction units; a single YFP adjacent to two CFPs contributes two units, so K_n is not generally the number of distinct fluorescent YFP molecules.

Table 1 summarizes the four models and the native ratio used for each.

Table 1 Overview of the four combinatorial fluorescence models

Model	Alphabet	Interaction	Native ratio
split-YFP	$\{A, B\}$	A–B fusion	$R_{SY} = 2E/n$
split-YFP + Sup35p	$\{A, B, S\}$	A–B fusion	$R_{SY,S} = 2E/n$
FRET YFP/CFP	$\{C, Y\}$	C–Y adjacency	$R_F = E/n$
FRET + Sup35p	$\{C, Y, S\}$	C–Y adjacency	$R_{F,S} = E/n$

The paper is organized as follows. Sections 2 and 3 treat split-YFP models, where the blocking constraint requires constrained-string enumeration and finite-state Markov chains. Sections 4 and 5 treat FRET models, where the absence of blocking makes the expectation a direct adjacent-pair count, although transfer matrices still encode the full distribution. Section 6 records finite- n variances and the finite-radius FRET expectation, and Section 7 gives limiting unequal-concentration formulas.

2 Analysis of exclusively split-YFP Amyloids

Let us first analyze amyloids consisting solely of YFP. YFP is a single polypeptide chain that can be split into two fragments, referred to as split-YFP. Call the distinct halves A and B. It is when these two halves A and B fuse that fluorescence occurs at

the point of contact. There are no restrictions on the patterns in the amyloid - one
can view all combinations of such halves via the regular expression $\{A, B\}^*$.

We are interested in positions of an amyloid where A and B come together and
fuse, producing yellow fluorescence. Such a scenario is essentially the reconstitution
of the full fluorophore structure of YFP. We refer to such points in the amyloid
stack as *fluorescing locations* (or *fluorescing sites*) throughout this paper. The same
terminology applies in Sections 4–5 to the FRET settings, where a fluorescing location
is a $\{C, Y\}$ adjacency at which energy transfer occurs.

Let n be the number of split-YFPs in a given amyloid. Clearly, there are 2^n possible
amyloids of length n .

We will start with the case where the halves of Split-YFP are well-mixed in a
neutral solution and of equal concentrations, and are thus equally likely to appear as
the next protein in an amyloid sequence.

2.1 Lighting Patterns and 01 representations

We abstract away from A and B specifically and instead analyze the “lighting
patterns” of amyloids of length n , defined as follows.

We ask: How many distinct patterns of fluorescing locations could we possibly see
from an amyloid of length n ?

We define the *01 representation* (or *01 pattern*) of an amyloid $s_1 s_2 \cdots s_n \in \{A, B\}^n$
as the binary string $f_1 f_2 \cdots f_n \in \{0, 1\}^n$ given by:

$$f_i = \begin{cases} 0, & \text{if } i = 1, \\ 1, & \text{if } i \geq 2, s_i \neq s_{i-1}, \text{ and } f_{i-1} = 0, \\ 0, & \text{otherwise.} \end{cases}$$

231 Informally, $f_i = 1$ indicates that a fusion occurs between positions $i - 1$ and i ,
 232 attributed to the later index i : the protein s_i differs from its neighbor s_{i-1} and that
 233 neighbor was not already consumed by a prior fusion. The first protein always receives
 234 $f_1 = 0$ since there is no predecessor. The blocking constraint ($f_{i-1} = 0$ required)
 235 ensures that both proteins in a fused pair are “used up,” preventing cascading fusions.
 236 Note that the 01 representation has the same length as the amyloid, and the mapping
 237 f is well-defined for any sequence in $\{A, B\}^n$.

242 The left-to-right, greedy pairing rule reflects an idealized sequential-polymerization
 243 picture: proteins are incorporated one at a time at a single growing end of the fibril,
 244 and each newly added protein either fuses with the current endpoint or does not.
 245 This greedy convention is motivated by the kinetics of sequential assembly—once a
 246 split-YFP pair reconstitutes, bimolecular fluorescence complementation is commonly
 247 treated as effectively irreversible on experimental timescales (Hu et al. 2002; Kerppola
 248 2006), so later arrivals cannot displace it in this null model. It should nevertheless be
 249 read as a modeling convention, not as a full kinetic model of fibril growth at multiple
 250 ends or with spatial rearrangements.

257 The convention is also compatible with a common static alternative at the level
 258 of counts. If one forms the path graph whose edges are adjacent unlike pairs and
 259 asks only for a maximum-cardinality matching, the leftmost greedy algorithm gives a
 260 maximum matching on a path: after taking the leftmost available edge, no maximum
 261 matching can use more than one edge among its two endpoints, and the remaining
 262 problem is again a shorter path. Thus the fluorescence count here agrees with the static
 263 maximum-matching count under unit weights, although the identity of the paired
 264 proteins remains tied to the chosen growth convention. Other static ensembles, such
 265 as Boltzmann-weighted matchings or geometry-dependent pairing propensities, would
 266 define different null models and are outside the scope of the present calculation.

For example, if we have *ABBAAAAABA*, then reading from left to right (because we have to think about how these proteins were added one at a time to analyze which ones fused), we get the 01 representation 0101000010. Writing these two representations side by side shows it:

ABBAAAAABA
0 10 10 00 01 0

We always start with a 0, since the first protein cannot cause the (non-existent) previous protein to fluoresce. Doing this, we no longer need think about whether we started with an A or a B.

As another example,

BBBBABABABABABABAAAAABAABBBBBABA
0 00 01 01 01 01 01 01 01 01 00 01 00 10 10 00 10 1

We observe that an unambiguous regular expression for these 01 patterns is $(0^*01)^*0^*$ (non-ambiguous in the sense required by the symbolic method: each string has a unique parse). Within the mathematical model, if the fluorescing locations were site-resolved, they would determine the 01 pattern: each fluorescing location corresponds to a 1.

We can count the total number of distinct 01 patterns of length n using the symbolic method for regular expressions (Flajolet and Sedgewick 2009, §I.4). Note that $(0^*01)^*0^* = (0^+1)^*0^+$: each block 0^*01 consists of zero or more leading zeros followed by 01, so the trailing 0 of 0^* and the leading 0 of 01 merge to give at least one zero before the 1, hence $0^*01 = 0^+1$. We assign weight x per symbol. Each block

323 $0^*01 = 0^+1$ has generating function $\frac{x}{1-x} \cdot x = \frac{x^2}{1-x}$, and the Kleene star $(0^*01)^*$ has
324 generating function $\frac{1}{1-x^2/(1-x)} = \frac{1-x}{1-x-x^2}$. Including the trailing 0^* with generating
325 function $\frac{1}{1-x}$, we obtain:
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$$G(x) := \sum_{n=0}^{\infty} a_n x^n = \frac{1-x}{1-x-x^2} \cdot \frac{1}{1-x} = \frac{1}{1-x-x^2}$$

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333 Since $(1-x-x^2) \sum_{n \geq 0} a_n x^n = 1$, we read off the recurrence $a_n = a_{n-1} + a_{n-2}$
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335 for $n \geq 2$, with $a_0 = a_1 = 1$.

336 We see that $a_0 = 1, a_1 = 1, a_2 = 2, a_3 = 3, a_4 = 5, a_5 = 8$, etc.

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338 So $a_n = F_{n+1}$, where F_n is the Fibonacci sequence with the standard convention
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340 $F_1 = F_2 = 1$. The coefficients $\binom{n-k}{k}$ appearing in this sum are precisely the entries
341 along the *shallow diagonals* of Pascal's triangle, and the identity $a_n = F_{n+1}$ is a
342 well-known consequence of summing along these diagonals. The weighted sums $S_1(n)$
343 and $S_2(n)$ evaluated in Section 2.4 can be viewed as weighted sums along these same
344 diagonals, with each entry scaled by $k \cdot 2^k$.
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352 2.2 Probability of k fluorescing locations in length n amyloid

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354 We now analyze the probability of seeing k fluorescing locations in an amyloid of
355 length n .
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361 2.2.1 Distinct 01 patterns with k 1s

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363 First, let us count the number of 01 patterns of length n with k 1s. Our invariant
364 states that no 1s may be adjacent, and the first digit must be a zero. Since $f_1 = 0$
365 always, the k ones must be placed among positions $2, 3, \dots, n$ (i.e., $n-1$ available
366 positions) with no two adjacent. This is equivalent to placing k non-adjacent objects
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in $n - 1$ boxes, giving:

$$N(n, k) := \binom{n-k}{k}, \quad k \leq \lfloor n/2 \rfloor$$

Indeed, the number of ways to place k non-adjacent objects in m boxes is $\binom{m-k+1}{k}$ (see (Stanley 2012, Exercise 1.34)); setting $m = n - 1$ gives $\binom{n-k}{k}$ as claimed.

2.2.2 Distinct amyloids with k 1s

Now, for $k \geq 1$, the number of distinct amyloids of length n with k fluorescing locations is:

$$\mathcal{A}(n, k) = 2^k \cdot N(n-2, k-1) + 2^{k+1} \cdot N(n-1, k) = 2^k \binom{n-k-1}{k-1} + 2^{k+1} \binom{n-k-1}{k}$$

The zero-fluorescence case is $\mathcal{A}(n, 0) = 2$, corresponding to the two constant amyloids $AA \cdots A$ and $BB \cdots B$. Throughout, binomial coefficients with lower index outside the usual range are interpreted as zero; nevertheless the displayed formula for $\mathcal{A}(n, k)$ is stated for $k \geq 1$, with $k = 0$ handled separately.

To justify the formula, fix an admissible 01 pattern f with k ones. Once s_1 is chosen, every later protein is determined except immediately after a 1 in the 01 pattern: if $f_i = 1$, then s_i is forced to be the opposite of s_{i-1} ; if $f_i = 0$ and $f_{i-1} = 0$, then s_i is forced to equal s_{i-1} ; if $f_i = 0$ and $f_{i-1} = 1$, then s_i is free because the previous protein has already been consumed by a fusion. Thus a fixed pattern ending in 0 has 2^{k+1} realizing amyloids (the initial choice, plus one free choice immediately after each of the k ones, all of which have successors), while a fixed pattern ending in 1 has 2^k realizing amyloids.

It remains only to count admissible 01 patterns by their final digit. Patterns ending in 0 are obtained by appending a zero to a length- $(n-1)$ admissible pattern with k ones, giving $N(n-1, k)$ choices. Patterns ending in 1 must have $f_{n-1} = 0$, and

415 deleting the final 01 leaves a length- $(n-2)$ admissible pattern with $k-1$ ones, giving
 416 $N(n-2, k-1)$ choices. Multiplying by the fixed-pattern counts proves the displayed
 417 formula.
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419 For example, when $k=5$ the two cases have the following degrees of freedom:

420 **01 pattern ends with a 0**

421 Consider the following pattern. We write the number of possible proteins (A or B)
 422 at that location below the corresponding location of the pattern.
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 428 000010000101001000010000
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 430 211112111121211211112111
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433 Indeed, for each block of 0's, at the beginning of the block, any of either A or B may
 434 appear there, since it will not fuse with the previous protein. Then thereafter each
 435 protein is uniquely determined by this choice through to the 1 that appears.
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437 Any pattern of this case therefore has 2^{k+1} possible amyloids associated with it.
 438 There are $N(n-1, k)$ possible patterns for this case.
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440 **01 pattern ends with a 1**

441 Consider the following 01 pattern. We write the number of possible proteins (A or
 442 B) at that location below the corresponding location of the 01 pattern.
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 448 000010000101001000000001
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 450 211112111121211211111111
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453 As we can see, putting a 1 at the end of the 01 pattern implies only 2^k possible
 454 amyloids with the particular pattern. There are $N(n-2, k-1)$ patterns with this
 455 property.
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Verification. For $n = 4$, $k = 1$: $\mathcal{A}(4, 1) = 2^1 \binom{2}{0} + 2^2 \binom{2}{1} = 2 + 8 = 10$. For $k = 2$:
 $\mathcal{A}(4, 2) = 2^2 \binom{1}{1} + 2^3 \binom{1}{2} = 4 + 0 = 4$ (the only 01 pattern is 0101, giving amyloids
ABAB, BABA, ABBA, BAAB). With $\mathcal{A}(4, 0) = 2$ (AAAA and BBBB), we have
 $2 + 10 + 4 = 16 = 2^4$. ✓

Hence the probability of seeing k fluorescing locations in an amyloid of length n is

$$P(n, k) = \frac{\mathcal{A}(n, k)}{2^n}$$

2.3 Expected number of fluorescing locations

This is simply the sum:

$$E(n) = \frac{1}{2^n} \sum_{k=1}^{\lfloor n/2 \rfloor} k \cdot \mathcal{A}(n, k) = \frac{1}{2^n} \sum_{k=1}^{\lfloor n/2 \rfloor} k \left(2^k \binom{n-k-1}{k-1} + 2^{k+1} \binom{n-k-1}{k} \right)$$

by definition of expected value.

2.4 Expected fraction of proteins involved in fluorescence

Since each fluorescing location involves two proteins, the fraction of proteins participating in fluorescence is:

$$R_{\text{SY}}(n) := \frac{2E(n)}{n} = \frac{1}{n \cdot 2^{n-1}} \sum_{k=1}^{\lfloor n/2 \rfloor} k \left(2^k \binom{n-k-1}{k-1} + 2^{k+1} \binom{n-k-1}{k} \right)$$

We claim that

$$\sum_{k=1}^{\lfloor n/2 \rfloor} k 2^k \binom{n-k-1}{k-1} = \frac{2^n(3n+2) + 2(6n-1)(-1)^n}{27} \quad (1)$$

and

$$\sum_{k=1}^{\lfloor n/2 \rfloor} k 2^k \binom{n-k-1}{k} = \frac{2^n(3n-4) - 2(3n-2)(-1)^n}{27} \quad (2)$$

We prove both identities using ordinary generating functions.

We will use the identity (see, e.g., (Wilf 2006, §1.2)) that for any fixed $j \geq 0$,

$$\sum_{n \geq 2j} \binom{n-j}{j} x^n = \frac{x^{2j}}{(1-x)^{j+1}} \quad (3)$$

which follows from the substitution $m = n - j$ and the known generating function

$$\sum_{m \geq j} \binom{m}{j} x^m = \frac{x^j}{(1-x)^{j+1}}.$$

Proof of (1). Let $S_1(n) := \sum_{k=1}^{\lfloor n/2 \rfloor} k 2^k \binom{n-k-1}{k-1}$. We derive the ordinary generating function $G_1(x) = \sum_{n \geq 0} S_1(n) x^n$.

Using the substitution $j = k - 1$,

$$S_1(n) = \sum_{j \geq 0} (j+1) 2^{j+1} \binom{n-j-2}{j}$$

Working as a formal power series, we may interchange the order of summation termwise. Applying (3) with a shift $n \mapsto n - 2$:

$$G_1(x) = \sum_{j \geq 0} (j+1) 2^{j+1} \cdot \frac{x^{2j+2}}{(1-x)^{j+1}} = \frac{2x^2}{1-x} \sum_{j \geq 0} (j+1) \left(\frac{2x^2}{1-x} \right)^j$$

Setting $u = 2x^2/(1-x)$ and using $\sum_{j \geq 0} (j+1) u^j = 1/(1-u)^2$, we have

$$G_1(x) = \frac{2x^2}{1-x} \cdot \frac{1}{(1-u)^2}$$

Since $1 - u = (1 - x - 2x^2)/(1 - x) = (1 - 2x)(1 + x)/(1 - x)$, this simplifies to

$$G_1(x) = \frac{2x^2(1 - x)}{(1 - 2x)^2(1 + x)^2}$$

We now perform a partial fraction decomposition:

$$G_1(x) = \frac{\alpha}{1 - 2x} + \frac{\beta}{(1 - 2x)^2} + \frac{\gamma}{1 + x} + \frac{\delta}{(1 + x)^2}$$

Multiplying both sides by $(1 - 2x)^2(1 + x)^2$ and evaluating at $x = 1/2$ gives $\beta = 1/9$, and at $x = -1$ gives $\delta = 4/9$. Expanding the right-hand side and comparing coefficients of x^0 and x^1 then yields $\alpha = -1/27$ and $\gamma = -14/27$. Now, using the standard power series

$$\begin{aligned} \frac{1}{1 - 2x} &= \sum_{n \geq 0} 2^n x^n, & \frac{1}{(1 - 2x)^2} &= \sum_{n \geq 0} (n + 1) 2^n x^n \\ \frac{1}{1 + x} &= \sum_{n \geq 0} (-1)^n x^n, & \frac{1}{(1 + x)^2} &= \sum_{n \geq 0} (n + 1) (-1)^n x^n \end{aligned}$$

we extract the coefficient of x^n :

$$\begin{aligned} S_1(n) &= -\frac{1}{27} \cdot 2^n + \frac{1}{9} (n + 1) 2^n - \frac{14}{27} (-1)^n + \frac{4}{9} (n + 1) (-1)^n \\ &= \frac{1}{27} [2^n (-1 + 3n + 3) + (-1)^n (-14 + 12n + 12)] \\ &= \frac{2^n (3n + 2) + 2(6n - 1) (-1)^n}{27} \end{aligned}$$

as claimed. $\square$

Proof of (2). Let $S_2(n) := \sum_{k=1}^{\lfloor n/2 \rfloor} k 2^k \binom{n-k-1}{k}$. As before, applying (3):

$$G_2(x) = \sum_{n \geq 0} S_2(n) x^n = \sum_{k \geq 1} k \cdot 2^k \cdot \frac{x^{2k+1}}{(1 - x)^{k+1}} = \frac{x}{1 - x} \cdot \frac{u}{(1 - u)^2}$$

599 where $u = 2x^2/(1-x)$ as before. This simplifies to

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$$G_2(x) = \frac{2x^3}{(1-2x)^2(1+x)^2}$$

Performing partial fractions, evaluating at $x = 1/2$ gives $\beta = 1/9$ and at $x = -1$ gives

$\delta = -2/9$. Comparing coefficients of x^0 and x^1 yields $\alpha = -7/27$ and $\gamma = 10/27$.

Extracting coefficients:

$$\begin{aligned} S_2(n) &= -\frac{7}{27} \cdot 2^n + \frac{1}{9}(n+1)2^n + \frac{10}{27}(-1)^n - \frac{2}{9}(n+1)(-1)^n \\ &= \frac{1}{27} [2^n(-7+3n+3) + (-1)^n(10-6n-6)] \\ &= \frac{2^n(3n-4) - 2(3n-2)(-1)^n}{27} \end{aligned}$$

as claimed. $\square$

Substituting (1) and (2):

$$R_{\text{SY}}(n) = \frac{1}{n \cdot 2^{n-1}} \sum_{k=1}^{\lfloor n/2 \rfloor} k \left(2^k \binom{n-k-1}{k-1} + 2^{k+1} \binom{n-k-1}{k} \right) \quad (4)$$

$$\begin{aligned} &= \frac{1}{n \cdot 2^{n-1}} \left(\frac{2^n(3n+2) + 2(6n-1)(-1)^n}{27} \right. \\ &\quad \left. + 2 \cdot \frac{2^n(3n-4) - 2(3n-2)(-1)^n}{27} \right) \quad (5) \end{aligned}$$

$$= \frac{1}{n \cdot 2^{n-1}} \cdot \frac{2^n(9n-6) + 6(-1)^n}{27} \quad (6)$$

$$= \frac{2(9n-6)}{27n} + \frac{6(-1)^n}{27n \cdot 2^{n-1}} \quad (7)$$

$$= \frac{2(3n-2)}{9n} + \frac{2(-1)^n}{9n \cdot 2^{n-1}} \quad (8)$$

Then as $n \rightarrow \infty$, the second term vanishes and we have

$$\lim_{n \rightarrow \infty} R_{\text{SY}}(n) = \lim_{n \rightarrow \infty} \frac{2(3n-2)}{9n} = \frac{6}{9} = \frac{2}{3} \quad (9)$$

2.4.1 Weighted shallow diagonal identities

The identities (1) and (2) have a natural interpretation in terms of Pascal's triangle. Recall that the entries along the shallow diagonals of Pascal's triangle are $\binom{n-k}{k}$ for $k = 0, 1, \dots, \lfloor n/2 \rfloor$, and their sum $\sum_{k=0}^{\lfloor n/2 \rfloor} \binom{n-k}{k} = F_{n+1}$ yields the Fibonacci numbers (Section 2). The identities (1) and (2) provide closed-form evaluations of *weighted* sums along these same diagonals, where each entry is scaled by $k \cdot 2^k$:

$$\sum_{k=1}^{\lfloor n/2 \rfloor} k \cdot 2^k \binom{n-k-1}{k-1} = \frac{2^n(3n+2) + 2(6n-1)(-1)^n}{27} \quad (10)$$

$$\sum_{k=1}^{\lfloor n/2 \rfloor} k \cdot 2^k \binom{n-k-1}{k} = \frac{2^n(3n-4) - 2(3n-2)(-1)^n}{27} \quad (11)$$

These identities arise here from the fluorescence analysis, but the sequences themselves are not new. Specifically, $S_1(n) = 2 \cdot \text{A127976}(n-1)$ and $S_2(n) = \text{A095977}(n-2)$, where OEIS lists A127976 as a convolution involving Jacobsthal numbers and A095977 as counting 2×2 tiles in tilings of $3 \times (n+1)$ rectangles (OEIS Foundation Inc.; OEIS Foundation Inc.). Both sequences satisfy the linear recurrence $a_n = 2a_{n-1} + 3a_{n-2} - 4a_{n-3} - 4a_{n-4}$ with different initial conditions, which can be verified from the generating function denominator $(1+x)^2(1-2x)^2 = 1 - 2x - 3x^2 + 4x^3 + 4x^4$. Direct computation confirms: $S_1(2) = 2 = 2 \cdot \text{A127976}(1)$ and $S_2(3) = 2 = \text{A095977}(1)$, with subsequent values matching by the shared recurrence.

The appearance of the same sequences in these different settings is explained by the shared reduction to constrained binary strings and rational generating functions

with denominator $(1+x)^2(1-2x)^2$. Thus the OEIS matches are useful checks and interpretations, but the main point here is their role in the fluorescence null model.

As a corollary, the linear combination $S_1(n) + 2S_2(n) = [2^n(9n-6) + 6(-1)^n]/27$ governs the expected fluorescence in split-YFP amyloids, connecting these combinatorial identities directly to the biophysical application.

2.4.2 Alternative derivation via Markov chain

We can also derive $E(n)$ directly by a Markov chain argument (Norris 1997), bypassing $\mathcal{A}(n, k)$ entirely. Consider building an amyloid one protein at a time, where each protein is independently A or B with equal probability 1/2. Define two states based on the rightmost protein:

- **State p :** the rightmost protein is *not* part of a fused pair.
- **State q :** the rightmost protein is part of a fused pair.

From state p (say the rightmost protein is A, unfused): with probability 1/2 the next protein is A (same type, no fusion, stay in p), and with probability 1/2 it is B (fusion, go to q , +1 fluorescence). By symmetry, the case for B is identical. So from p : transition to p with probability 1/2 and to q with probability 1/2, with expected fluorescence 1/2.

From state q (rightmost protein is fused): the next protein is A or B each with probability 1/2, and in either case goes to state p with no fluorescence (since the current protein is already fused). So from q : transition to p with probability 1.

With $p_1 = 1$ (the first protein is always unfused), the recurrence is

$$p_{n+1} = \underbrace{\frac{1}{2} p_n}_{\text{stay in } p} + \underbrace{(1 - p_n)}_{\text{from } q \text{ to } p} = 1 - \frac{1}{2} p_n$$

which has fixed point $p^* = 2/3$ and general solution

$$p_n = \frac{2}{3} + \frac{1}{3} \left(-\frac{1}{2} \right)^{n-1}$$

The expected fluorescence contributed by the $(n+1)$ th protein is $e_{n+1} = \frac{1}{2}p_n$, so

$$\begin{aligned} E(n) &= \sum_{i=2}^n e_i = \frac{1}{2} \sum_{j=1}^{n-1} p_j = \frac{1}{2} \sum_{j=1}^{n-1} \left[\frac{2}{3} + \frac{1}{3} \left(-\frac{1}{2} \right)^{j-1} \right] \\ &= \frac{n-1}{3} + \frac{1}{6} \cdot \frac{1 - (-1/2)^{n-1}}{3/2} = \frac{n-1}{3} + \frac{1}{9} \left(1 - \left(-\frac{1}{2} \right)^{n-1} \right) \end{aligned}$$

Therefore

$$R_{\text{SY}}(n) = \frac{2E(n)}{n} = \frac{2(n-1)}{3n} + \frac{2}{9n} \left(1 - \left(-\frac{1}{2} \right)^{n-1} \right) \longrightarrow \frac{2}{3} \text{ as } n \rightarrow \infty$$

Since this must agree with the combinatorial derivation, equating the two expressions for $E(n)$ provides an independent verification of the sum identities (1) and (2). Specifically,

$$\frac{S_1(n) + 2S_2(n)}{2^n} = \frac{n-1}{3} + \frac{1}{9} \left(1 - \left(-\frac{1}{2} \right)^{n-1} \right)$$

To verify this, substitute the closed forms from (1) and (2):

$$\frac{S_1(n) + 2S_2(n)}{2^n} = \frac{2^n(9n-6) + 6(-1)^n}{27 \cdot 2^n} = \frac{3n-2}{9} + \frac{2(-1)^n}{9 \cdot 2^n}$$

while the Markov chain gives $E(n) = \frac{n-1}{3} + \frac{1}{9} - \frac{(-1)^{n-1}}{9 \cdot 2^{n-1}} = \frac{3n-2}{9} + \frac{2(-1)^n}{9 \cdot 2^n}$, confirming the two expressions are identical.

We can also verify the above via computer simulation (see `split_yfp_simulation.py`; all scripts are available at <https://github.com/>)

[danielchen0/amyloids](#)). A sample of computed values is shown in Table 2, and the first 400 values appear in Figure 1.

n	$R_{SY}(n)$	n	$R_{SY}(n)$	n	$R_{SY}(n)$
2	0.500000	10	0.622266	20	0.644444
3	0.500000	11	0.626243	25	0.648889
4	0.562500	12	0.629639	50	0.657778
5	0.575000	13	0.632474	100	0.662222
6	0.593750	14	0.634923	200	0.664444
7	0.602679	15	0.637036	400	0.665556

$R_{SY}(n) \rightarrow 2/3 \approx 0.666667$

Table 2 Sample values of $R_{SY}(n)$ for split-YFP amyloids.

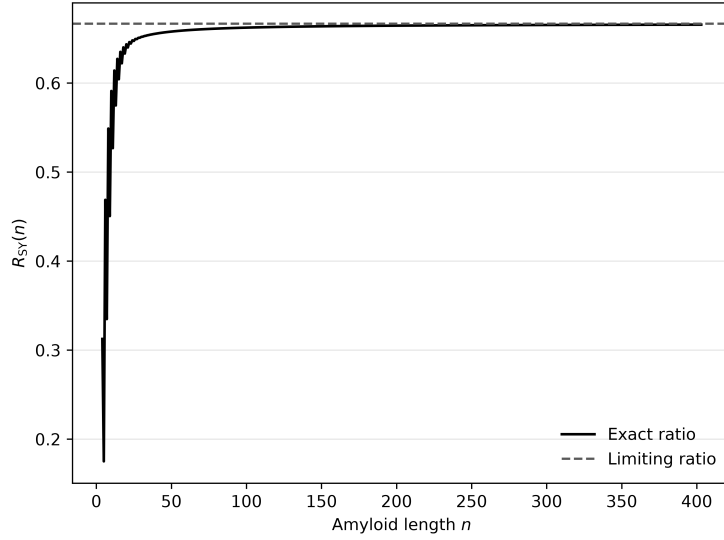


Fig. 1 $R_{SY}(n)$ for split-YFP amyloids. The dashed horizontal line is $2/3$ and the solid curve is the exact $R_{SY}(n)$. Convergence to the limiting ratio is algebraic ($O(1/n)$), with a smaller alternating exponential correction.

3 Introducing Sup35p into Amyloids

We now want to consider the facts above but with the introduction of yeast prion protein Sup35p (which we label with an S) into our amyloids ([Patino et al. 1996](#)).

In this null model, Sup35p is used as an abstract inert third species: we do not model its own prion biology, only its role as a non-fluorescent stack partner. It cannot participate in any fluorescent interaction, and its sole effect in the model is to sterically separate adjacent split-YFP halves, preventing them from conjoining and fluorescing. Our amyloids can now be described using the regular expression $\{A, B, S\}^*$. We would like to first analyze the case where Sup35p and the halves of Split-YFP come in equal concentrations, and are thus equally likely to occur as the next protein in an amyloid sequence.

The 01 representation introduced in Section 2 must be extended to $\{A, B, S\}^n$ to reflect Sup35p's inertness. Applied verbatim, the Section 2 rule " $f_i = 1$ iff $s_i \neq s_{i-1}$ and $f_{i-1} = 0$ " would mark an $S \rightarrow A$ adjacency as fluorescing, contradicting the modeling assumption that S blocks fluorescence. The intended rule, used implicitly throughout this section, is:

$$f_i = \begin{cases} 0, & \text{if } i = 1, \\ 1, & \text{if } i \geq 2, \{s_{i-1}, s_i\} = \{A, B\}, \text{ and } f_{i-1} = 0, \\ 0, & \text{otherwise.} \end{cases}$$

That is, a 1 requires both proteins in the adjacency to be split-YFP halves (one A and one B), and that the left protein not already be consumed by a prior fusion. With this extension, the set of 01 patterns produced by $\{A, B, S\}^n$ sequences is exactly the same as in Section 2: the regular expression $(0^*01)^*0^*$ still describes all attainable patterns, since insertions of S only stretch existing zero runs.

875 3.1 Number of distinct lighting patterns and 01 representations

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877 The introduction of Sup35p does not change this. When we abstract away the details
878 we are still looking at 01 patterns of the form $(0^*01)^*0^*$. Hence we have F_{n+1} distinct
880 lighting patterns for an amyloid of length n (using the convention $F_1 = F_2 = 1$).
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883 3.2 Probability of k fluorescing locations in length n amyloid

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885 We derive the joint distribution of length and fluorescence count from a bivariate
886 generating function obtained by a two-state Markov decomposition, and then unpack
887 it into a block-decomposition formula that mirrors Section 2.
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891 3.2.1 Bivariate generating function

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893 Throughout the transfer-matrix derivations, $M(w)_{ij}$ weights a step from state i to
894 state j , row vectors multiply on the left, and a final column vector of ones sums over
895 terminal states. Build an A-B-S amyloid one protein at a time, classifying its rightmost
896 protein into two lumped states:
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- p : rightmost is A or B and is not currently part of a fused pair.
- q : rightmost is either S, or is A/B that has just been fused with its left neighbor.

The transitions, weighted by z for length and w for each fluorescing site produced, are:

$$M(w) = \begin{pmatrix} 1 & 1+w \\ 2 & 1 \end{pmatrix},$$

where the rows index the from-state (p, q) and the columns the to-state. From p (say rightmost is A, unfused): next protein A yields one new amyloid ending in p (weight 1); next protein B fuses and yields one amyloid ending in q (weight w , +1 fluorescence); next protein S yields one amyloid ending in q (weight 1). From q : either A or B sends us to p (weight 2 for two letters), and S sends us back to q (weight 1). This two-state

lumping is valid here because A and B have equal weights and symmetric transition behavior; for unequal A/B concentrations the unfused A and unfused B states must be separated, as in Section 7. For $n \geq 1$, the row vector of length- n weighted counts is $(2, 1)M(w)^{n-1}$; the initial vector $(2, 1)$ records the two one-letter amyloids A/B in state p and the one one-letter amyloid S in state q . Hence the joint generating function

$$F(z, w) := \sum_{n \geq 0} \sum_{k \geq 0} \mathcal{A}(n, k) w^k z^n$$

(where in this section $\mathcal{A}(n, k)$ counts amyloids over $\{A, B, S\}$, and $\mathcal{A}(0, 0) := 1$ by convention) is

$$F = 1 + \sum_{n \geq 1} z^n (2, 1) M(w)^{n-1} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = 1 + z(2, 1)(I - zM(w))^{-1} \begin{pmatrix} 1 \\ 1 \end{pmatrix},$$

which evaluates to

$$F(z, w) = \frac{1 + z}{1 - 2z - (1 + 2w)z^2}. \quad (12)$$

Equivalently, the coefficients satisfy the recurrence $\mathcal{A}(n, k) = 2\mathcal{A}(n - 1, k) + \mathcal{A}(n - 2, k) + 2\mathcal{A}(n - 2, k - 1)$ with the appropriate small- n initial conditions. Two specializations recover familiar facts. At $w = 0$, we count amyloids with no fluorescence: $F(z, 0) = (1 + z)/(1 - 2z - z^2)$, whose coefficients b_n enumerate A-B-S sequences with all-zero 01 representation. At $w = 1$, all amyloids are counted: $F(z, 1) = (1 + z)/(1 - 2z - 3z^2) = 1/(1 - 3z)$ (after canceling the factor $1 + z$ from $1 - 2z - 3z^2 = (1 - 3z)(1 + z)$), so $[z^n]F(z, 1) = 3^n$ as required.

The expected value $E(n)$ can now be extracted in a few lines from $\partial F / \partial w|_{w=1}$:

$$\left. \frac{\partial F}{\partial w} \right|_{w=1} = \frac{2z^2(1 + z)}{(1 - 2z - 3z^2)^2} = \frac{2z^2}{(1 - 3z)^2(1 + z)},$$

and partial-fraction expansion (carried out explicitly in Section 3.4 below as a consistency check) yields the closed form

$$E(n) \cdot 3^n = [z^n] \frac{2z^2}{(1-3z)^2(1+z)} = \frac{3^{n-1}(4n-3) + (-1)^n}{8}.$$

Equivalently, $E(n) = (4n-3)/24 + (-1)^n/(8 \cdot 3^n) = (4n-3)/24 - (-1)^{n-1}/(24 \cdot 3^{n-1})$, which is the same formula derived by the explicit Markov recurrence in Section 3.4.

3.2.2 Block decomposition

The bivariate generating function (12) admits a more explicit reading via the regular-expression decomposition $(0^*01)^*0^*$ of the 01 pattern. Each amyloid of length n with k ones in its 01 pattern decomposes uniquely as k consecutive “active blocks” of the form 0^+1 , followed by a trailing run of zeros:

$$\underbrace{0 \cdots 0 1}_{\phi_1} \underbrace{0 \cdots 0 1}_{\phi_2} \cdots \underbrace{0 \cdots 0 1}_{\phi_k} \underbrace{0 \cdots 0}_q$$

with $\phi_i \geq 2$ and $q \geq 0$. The protein choices within different blocks are independent: once block i ends with a fusion at position j (so s_j is consumed by the previous fusion), the protein at position $j+1$ may be any of $\{A, B, S\}$ unconstrained, since $f_{j+1} = 0$ is forced regardless of s_{j+1} . Hence the number of amyloids realizing a given block decomposition factors:

$$\mathcal{A}(n, k) = \sum_{q=0}^{n-2k} b_q \sum_{\substack{\phi_1 + \cdots + \phi_k = n-q \\ \phi_i \geq 2}} \prod_{i=1}^k a_{\phi_i}, \quad (13)$$

where a_ϕ counts A-B-S patterns of length ϕ whose 01 representation is $0^{\phi-1}1$, and b_q counts patterns of length q with all-zero 01 representation. For $k=0$, the formula reduces to the all-zero case $\mathcal{A}(n, 0) = b_n$. Thus $P(n, k) = \mathcal{A}(n, k)/3^n$. Equation (13)

is the convolutional unfolding of (12): writing $F(z, w) = B(z)/(1 - wA(z))$ with $A(z) = \sum_{\phi \geq 2} a_\phi z^\phi = 2z^2/(1 - 2z - z^2)$ and $B(z) = \sum_{q \geq 0} b_q z^q = (1 + z)/(1 - 2z - z^2)$, one verifies

$$\frac{B(z)}{1 - wA(z)} = \frac{(1 + z)/(1 - 2z - z^2)}{1 - 2wz^2/(1 - 2z - z^2)} = \frac{1 + z}{1 - 2z - (1 + 2w)z^2},$$

recovering (12).

Unlike the two-letter split-YFP case, this gives $P(n, k)$ most naturally as a coefficient or convolution formula rather than as a single binomial closed form; the rational bivariate generating function is the compact representation of the full distribution.

3.2.3 The sequences a_n and b_n

Both sequences are governed by the same denominator $1 - 2x - x^2$. From the regular expressions $S^*\{AA^*SS^*, BB^*SS^*\}^*\{AA^*, BB^*, \varepsilon\}$ for the b -sequence and $S^*\{AA^*SS^*, BB^*SS^*\}^*\{AA^*B, BB^*A\}$ for the a -sequence (setting $A = B = S = x$ throughout), one obtains the ordinary generating functions

$$\sum_{n \geq 0} b_n x^n = \frac{1 + x}{1 - 2x - x^2}, \quad \sum_{n \geq 0} a_n x^n = \frac{2x^2}{1 - 2x - x^2},$$

yielding the recurrences $b_n = 2b_{n-1} + b_{n-2}$ for $n \geq 2$ (with $b_0 = 1, b_1 = 3$) and $a_n = 2a_{n-1} + a_{n-2}$ for $n \geq 3$ (with $a_0 = a_1 = 0, a_2 = 2$). The denominator's roots are reciprocals of $1 \pm \sqrt{2}$, and clean partial-fraction decomposition gives the closed forms

$$b_n = \frac{(1 + \sqrt{2})^{n+1} + (1 - \sqrt{2})^{n+1}}{2}, \quad n \geq 0, \quad (14)$$

$$a_n = \left(1 + \frac{\sqrt{2}}{2}\right)(1 - \sqrt{2})^n + \left(1 - \frac{\sqrt{2}}{2}\right)(1 + \sqrt{2})^n, \quad n \geq 1 \quad (15)$$

(the formula for a_n does not extend to $n = 0$, where it gives 2 rather than $a_0 = 0$; this is absorbed by the non-homogeneous initial conditions). Both sequences appear in OEIS: $b_n = \text{A001333}(n+1)$ and $a_n/2 = \text{A000129}(n-1)$ for $n \geq 1$, where A001333 and A000129 are respectively the continued-fraction numerators of $\sqrt{2}$ and the Pell numbers (OEIS Foundation Inc.; OEIS Foundation Inc.). Verification: $a_2 = (1 + \sqrt{2}/2)(3 - 2\sqrt{2}) + (1 - \sqrt{2}/2)(3 + 2\sqrt{2}) = 6 - (\sqrt{2}/2)(4\sqrt{2}) = 6 - 4 = 2$; $b_0 = (1 + \sqrt{2} + 1 - \sqrt{2})/2 = 1$. The first values are $b_0, b_1, \dots = 1, 3, 7, 17, 41, 99, \dots$ and $a_0, a_1, \dots = 0, 0, 2, 4, 10, 24, \dots$.

Worked example. For $n = 4$, $k = 1$, the block decomposition has one block of length $\phi_1 \geq 2$ followed by a tail of length $q = 4 - \phi_1$; the three cases are shown in Table 3. So $\mathcal{A}(4, 1) = 14 + 12 + 10 = 36$, confirmed by brute-force enumeration. Cross-

Table 3 Block-decomposition cases for $n = 4$ and $k = 1$

ϕ_1	q	01 pattern	$a_{\phi_1} \cdot b_q$
2	2	0100	$2 \cdot 7 = 14$
3	1	0010	$4 \cdot 3 = 12$
4	0	0001	$10 \cdot 1 = 10$

checking against (12): the coefficient of $z^4 w^1$ in $F(z, w)$, computed via the recurrence, equals 36 as well.

3.3 Expected number of fluorescing locations

This is simply the sum:

$$E(n) = \frac{1}{3^n} \sum_{k=1}^{\lfloor n/2 \rfloor} k \cdot \mathcal{A}(n, k)$$

by definition of expected value.

3.4 Expected fraction of proteins involved in fluorescence

This is the number:

$$R_{SY,S}(n) = \frac{2E(n)}{n}$$

We prove that $\lim_{n \rightarrow \infty} R_{SY,S}(n) = 1/3$ by computing $E(n)$ directly using a Markov chain argument, bypassing the combinatorial formula for $\mathcal{A}(n, k)$.

Consider building an amyloid one protein at a time, where each protein is independently A, B, or S with equal probability $1/3$. We define two states based on the status of the most recently added protein:

- **State p :** the rightmost protein is A or B and is *not* part of a fused pair (and so is available to fuse with the next protein).
- **State q :** the rightmost protein is either S, or is A/B but already part of a fused pair (and so the next protein cannot fuse with it).

The transition probabilities are as follows. From state p (say the rightmost protein is A, unfused):

- With probability $1/3$, the next protein is A (same type): no fusion, go to p .
- With probability $1/3$, the next protein is B (different type): fusion occurs (+1 fluorescence), go to q .
- With probability $1/3$, the next protein is S: no fusion, go to q .

By symmetry between A and B, the case where the rightmost is B is identical. So from p : transition to p with probability $1/3$, to q with probability $2/3$, with expected fluorescence $1/3$.

From state q , the rightmost protein cannot fuse regardless of the next protein's type. The next protein is A, B, or S each with probability $1/3$, and is always unfused. So from q : transition to p with probability $2/3$ (next is A or B), to q with probability $1/3$ (next is S), with expected fluorescence 0.

Let p_n denote the probability of being in state p after the n th protein. The initial state is p with probability $2/3$ and q with probability $1/3$, so $p_1 = 2/3$. The recurrence is:

$$p_{n+1} = \frac{1}{3}p_n + \frac{2}{3}(1 - p_n) = \frac{2}{3} - \frac{1}{3}p_n$$

This has fixed point $p^* = 1/2$, and the general solution is

$$p_n = \frac{1}{2} + \frac{(-1/3)^{n-1}}{6}$$

which is verified by checking $p_1 = 1/2 + 1/6 = 2/3$.

The expected fluorescence contributed by the $(n+1)$ th protein is $e_{n+1} = p_n/3$, so

$$\begin{aligned} E(n) &= \sum_{i=2}^n e_i = \frac{1}{3} \sum_{j=1}^{n-1} p_j = \frac{1}{3} \sum_{j=1}^{n-1} \left[\frac{1}{2} + \frac{(-1/3)^{j-1}}{6} \right] \\ &= \frac{n-1}{6} + \frac{1}{18} \cdot \frac{1 - (-1/3)^{n-1}}{4/3} = \frac{n-1}{6} + \frac{1}{24} (1 - (-1/3)^{n-1}) \\ &= \frac{4n-3}{24} - \frac{(-1)^{n-1}}{24 \cdot 3^{n-1}} \end{aligned}$$

Therefore

$$R_{SY,S}(n) = \frac{2E(n)}{n} = \frac{4n-3}{12n} + \frac{(-1)^n}{12n \cdot 3^{n-1}}$$

and as $n \rightarrow \infty$, the second term vanishes:

$$\lim_{n \rightarrow \infty} R_{SY,S}(n) = \lim_{n \rightarrow \infty} \frac{4n-3}{12n} = \frac{4}{12} = \frac{1}{3}$$

The block decomposition (13) and the transfer-matrix generating function (12) give two views of the same coefficients $\mathcal{A}(n, k)$. Taking expectations then gives the identity

$$\frac{1}{3^n} \sum_{k=1}^{\lfloor n/2 \rfloor} k \cdot \mathcal{A}(n, k) = \frac{4n-3}{24} - \frac{(-1)^{n-1}}{24 \cdot 3^{n-1}}$$

where $\mathcal{A}(n, k)$ is the combinatorial formula involving the a_{ϕ_i} and b_q sequences with $\sqrt{2}$. The left-hand side is the definition of $E(n)$, and the right-hand side is the closed form obtained from the transfer matrix or, equivalently, from the explicit Markov recurrence above. This is best read as a consistency check between the coefficient-level distribution and the expectation-level calculation, rather than as two wholly independent Markov-free derivations. Numerically, at $n = 4$, the left-hand side gives $(1 \cdot 36 + 2 \cdot 4)/81 = 44/81$, and the right-hand side gives $(4 \cdot 4 - 3)/24 + 1/(24 \cdot 27) = 13/24 + 1/648 = 44/81$. $\checkmark$

A simulation (see `sup35_simulation.py`) confirms this. A sample of computed values is shown in Table 4, and the graph appears in Figure 2.

n	$R_{\text{SY},S}(n)$	n	$R_{\text{SY},S}(n)$	n	$R_{\text{SY},S}(n)$
2	0.222222	10	0.308334	20	0.320833
3	0.246914	11	0.310606	25	0.323333
4	0.271605	12	0.312500	50	0.328333
5	0.283128	13	0.314103	100	0.330833
6	0.291724	14	0.315476	200	0.332083
7	0.297603	15	0.316667	400	0.332708

$$R_{\text{SY},S}(n) \rightarrow 1/3 \approx 0.333333$$

Table 4 Sample values of $R_{\text{SY},S}(n)$ for split-YFP amyloids with Sup35p.

4 Using FRET in amyloids of YFP and CFP

Consider the case where intact YFP and CFP are well-mixed in a container and observed under excitation of the CFP donor. When CFP and YFP are in close spatial proximity, nonradiative energy transfer from excited CFP to the YFP acceptor can lead to yellow YFP emission through FRET (Förster 1948; Patterson et al. 2000; Lakowicz 2006).

We are interested in measuring the *total* yellow fluorescence emanating from amyloids of this kind. In the counting convention used here, an adjacent CFP–YFP pair

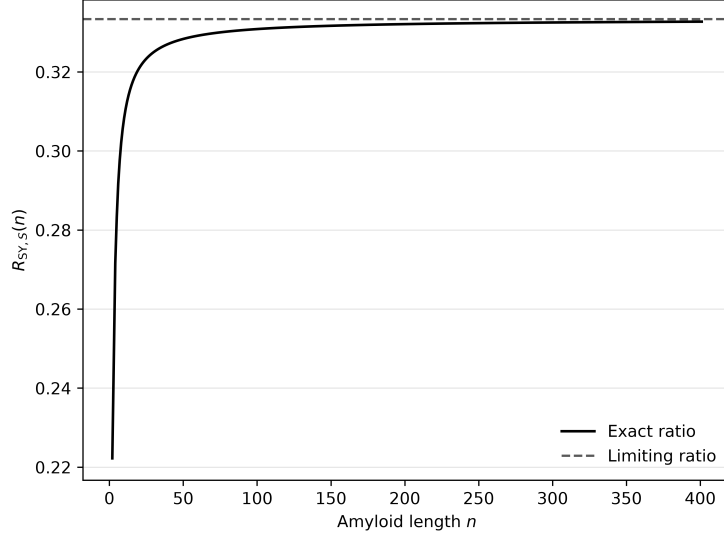


Fig. 2 $R_{SY,S}(n)$ for split-YFP amyloids with Sup35p. The dashed horizontal line is $1/3$ and the solid curve is the exact $R_{SY,S}(n)$. Convergence to the limiting ratio is algebraic ($O(1/n)$), with a smaller alternating exponential correction.

contributes one unit of yellow fluorescence; a YFP sandwiched between two CFPs therefore contributes two units. This should be read as an additive, time-averaged null-model convention rather than a photophysical prediction about a single acceptor molecule. In practice, real multi-donor FRET to one acceptor does not add linearly in this simple way: efficiencies depend on distance, orientation, donor competition, and saturation effects. We adopt the unit-additive convention for analytical tractability. A more realistic model incorporating geometry-dependent FRET efficiencies would require replacing the simple adjacent-pair counting argument below with a specified physical contribution model.

Label YFP as Y and CFP as C. The blocking constraint $f_{i-1} = 0$ that was central to Sections 2 and 3 does *not* apply here: by our additive convention, both CFP neighbors of a YFP may contribute to the total count, so consecutive 1s in the 01 representation are allowed. In the FRET sections, the 01 pattern is accordingly

defined by

$$f_i = \begin{cases} 0, & \text{if } i = 1, \\ 1, & \text{if } i \geq 2 \text{ and } \{s_{i-1}, s_i\} = \{C, Y\}, \\ 0, & \text{otherwise,} \end{cases}$$

with no constraint on f_{i-1} . The protein words are elements of $\{C, Y\}^n$, while the associated 01 patterns are exactly the binary strings of length n whose first symbol is 0. Indeed, after choosing the first protein, each subsequent bit determines whether the next protein is equal to or different from the previous one: a 1 forces it to differ, and a 0 forces it to match. Thus there are 2^{n-1} possible 01 patterns, and each is realized by exactly two Y-C sequences, according to the choice of the first protein.

The number of 1's in the 01 pattern is exactly the total yellow fluorescence in this additive convention: a 1 appears precisely when an adjacent CFP-YFP pair contributes one unit of FRET fluorescence. For example:

$$\begin{aligned} \{C, Y\}^*: & \text{ CYYYCYCYCC} \\ (0^*01^*1)^*0^*: & 0 \ 10 \ 01 \ 11 \ 10 \ 10 \\ \text{Yellow fluorescence:} & 0 \ 10 \ 10 \ 20 \ 11 \ 00 \end{aligned}$$

Note that in the 01 pattern, each 1 is attributed to the right endpoint of a $\{C, Y\}$ adjacency as a bookkeeping convention, not to the physically emitting protein. The key claim is that the *total* number of 1s equals the total yellow fluorescence, which we verify: both rows sum to 6.

4.1 Probability of k total yellow fluorescence in length n amyloid

The number of amyloids of length n with k total yellow fluorescence is:

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$$\mathcal{A}(n, k) = 2 \binom{n-1}{k}, \quad k \in [0, n-1]$$

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1339 and thus the probability of observing k total yellow fluorescence in a length n

1340 amyloid is

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$$P(n, k) = \frac{2 \binom{n-1}{k}}{2^n} = \frac{\binom{n-1}{k}}{2^{n-1}}, \quad k \in [0, n-1]$$

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1348 This is the Binomial($n-1, 1/2$) distribution, which can also be seen directly. Let

1349 $\varepsilon_i = \pm 1$ according to whether protein i is C or Y, and define $d_i = \varepsilon_i \cdot \varepsilon_{i-1}$ for $i \geq 2$.

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1351 Then $f_i = 1$ iff $d_i = -1$. The map $(\varepsilon_1, \dots, \varepsilon_n) \mapsto (\varepsilon_1, d_2, \dots, d_n)$ is a bijection on

1352

1353 $\{\pm 1\}^n$ (since $\varepsilon_i = \varepsilon_1 \prod_{j=2}^i d_j$ is recoverable), so $(d_2, \dots, d_n)$ are mutually indepen-

1354 dent and uniform on $\{\pm 1\}$. Hence $f_2, \dots, f_n$ are mutually independent Bernoulli($1/2$)

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1356 despite their overlapping definition, and their sum is Binomial($n-1, 1/2$).

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4.2 Expected total yellow fluorescence

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1363 This is simply the sum:

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$$E(n) = \sum_{k=0}^{n-1} k \cdot P(n, k)$$

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by definition of expected value.

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Note that, by a standard identity ([Graham et al. 1994](#), §5.1, Eq. 5.5):

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$$E(n) := \frac{1}{2^{n-1}} \cdot \sum_{k=0}^{n-1} k \cdot \binom{n-1}{k} = \frac{1}{2^{n-1}} \cdot (n-1)2^{n-2} = \frac{n-1}{2}$$

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So that for very large n , we expect to see about as much yellow unit fluorescence

as half the length of the amyloid.

Note that $E(n) = (n - 1)/2$ also follows directly from linearity of expectation: the total yellow fluorescence equals the number of adjacent $\{C, Y\}$ pairs, and there are $n - 1$ adjacent pairs each with probability $P(\{C, Y\}) = 2 \cdot (1/2)^2 = 1/2$ of being a $\{C, Y\}$ pair (since FRET interactions, unlike split-YFP fusion, have no blocking constraint). We define the fluorescence ratio as

$$R_F(n) := E(n)/n = \frac{n - 1}{2n} \longrightarrow \frac{1}{2} \text{ as } n \rightarrow \infty$$

The first 400 values of $R_F(n)$ appear in Figure 3.

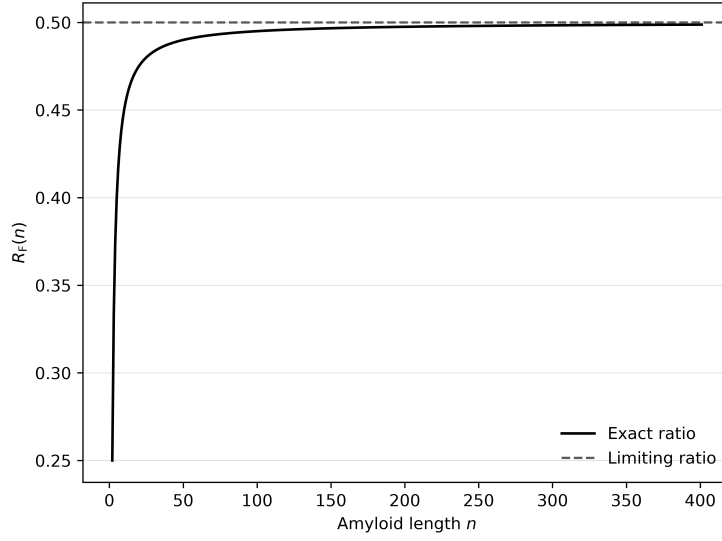


Fig. 3 $R_F(n)$ for FRET YFP/CFP amyloids. The dashed horizontal line is $1/2$ and the solid curve is the exact $R_F(n) = (n - 1)/(2n)$. Convergence is algebraic ($O(1/n)$), with no exponential correction.

5 Using FRET in amyloids of YFP, CFP, and Sup35p

As with Section 3, we will now analyze the case of introducing Sup35p to the system in Section 4. Thus we have Sup35p, YFP, and CFP well mixed and of equal concentrations in a container under CFP donor excitation. An adjacent CFP–YFP pair

1427 contributes one unit of FRET fluorescence in the additive convention of Section 4;
1428
1429 a YFP sandwiched between two CFPs contributes two such units. The presence of
1430 Sup35p blocks CFP–YFP adjacency in the one-dimensional model by separating the
1431
1432 fluorescent proteins.

1433 Our protein words are elements of $\{C, Y, S\}^*$, where C represents CFP, Y rep-
1434
1435 resents YFP, and S represents Sup35p. The 01 representation uses the same local
1436
1437 FRET rule as Section 4:

$$f_i = \begin{cases} 0, & \text{if } i = 1, \\ 1, & \text{if } i \geq 2 \text{ and } \{s_{i-1}, s_i\} = \{C, Y\}, \\ 0, & \text{otherwise.} \end{cases}$$

1446
1447 Thus every adjacency involving S contributes 0. As in Section 4, every binary string
1448
1449 with first symbol 0 is attainable, since the C-Y-only words already realize all such
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1451 patterns. Unlike the two-letter case, however, a fixed 01 pattern no longer determines
1452 the protein word up to the first letter, because zeros may arise from equal C/Y
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1454 neighbors or from the presence of S.

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1457 5.1 Probability of k total yellow fluorescence in length n

1458 amyloid

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1461 The block-decomposition approach used for Section 3.2 does not transfer cleanly to
1462
1463 this setting. The reason is that, unlike split-YFP fusion, FRET allows consecutive 1s
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1465 in the 01 pattern (a YFP sandwiched between two CFPs fluoresces twice), so each
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1467 block has the form 0^+1^+ rather than 0^+1 . The last protein of a 1^+ run is always in
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1469 $\{C, Y\}$, and the first protein of the next block must form $f = 0$ with it—hence must
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1471 match it or be S, only 2 choices among $\{C, Y, S\}$. By contrast, in Section 3.2, the
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1473 protein after a 1 is already blocked from fusing leftward, so the next protein is free

among all three letters. This boundary coupling means a FRET block decomposition would need to carry boundary-state information; the transfer matrix is the simpler way to track the rightmost protein type.

5.1.1 Bivariate generating function

Lump $\{C, Y\}$ into a single state (justified by the $C \leftrightarrow Y$ symmetry of the model under equal concentrations), and track two states by the rightmost protein type. Without this symmetry, for example if $p_C \neq p_Y$, this lumping would need to be replaced by a chain tracking C-ending and Y-ending states separately.

- p : rightmost is C or Y.
- q : rightmost is S.

The weighted transition matrix (rows = from, cols = to; z marks length, w marks each fluorescing site) is

$$M(w) = \begin{pmatrix} 1 + w & 1 \\ 2 & 1 \end{pmatrix}.$$

From p (say rightmost is C): next protein C stays in p with no fluorescence (weight 1); next protein Y stays in p but produces a fluorescence (weight w); next protein S goes to q (weight 1). By $C \leftrightarrow Y$ symmetry the rightmost-Y case is symmetric. From q (rightmost is S): C and Y carry no fluorescence with S, so next $\in \{C, Y\}$ takes us to p (combined weight 2), and next = S stays in q (weight 1). For $n \geq 1$, the row vector of length- n weighted counts is $(2, 1)M(w)^{n-1}$; again, $(2, 1)$ counts the two possible first letters C/Y in state p and the single first letter S in state q . Thus the joint generating function

$$F_{F,S}(z, w) := \sum_{n \geq 0} \sum_{k \geq 0} \mathcal{A}(n, k) w^k z^n$$

1519 (with $\mathcal{A}(0,0) := 1$) is $1 + z(2,1)(I - zM(w))^{-1}\binom{1}{1}$, namely

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$$F_{F,S}(z, w) = \frac{1 + (1 - w)z}{1 - (2 + w)z - (1 - w)z^2}. \quad (16)$$

1525

1526 Equivalently, the coefficients satisfy the recurrence $f_n = (2 + w)f_{n-1} + (1 - w)f_{n-2}$

1527 with $f_0 = 1$, $f_1 = 3$. At $w = 0$ this recovers $F_{F,S}(z, 0) = (1 + z)/(1 - 2z - z^2)$,

1528

1529 the same denominator as b_n in Section 3.2—which is correct, since both sequences

1530

1531 count length- n A-B-S (or C-Y-S) strings with no fluorescent adjacency, regardless of

1532 alphabet relabeling. At $w = 1$ the generating function becomes $F_{F,S}(z, 1) = 1/(1 - 3z)$,

1533

1534 so $[z^n]F_{F,S}(z, 1) = 3^n$ as required.

1535

1536 The denominator factors as $(1 - \lambda_+z)(1 - \lambda_-z)$ where the eigenvalues of $M(w)$ are

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$$\lambda_{\pm}(w) = 1 + \frac{w \pm \sqrt{w^2 + 8}}{2},$$

1542 recovering $\lambda_+(1) = 3$, $\lambda_-(1) = 0$, and $\lambda_{\pm}(0) = 1 \pm \sqrt{2}$ consistently with the $w = 0, 1$

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1544 specializations.

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1550 5.1.2 Expected value via $\partial F_{F,S}/\partial w$

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1552 Differentiating (16) in w and evaluating at $w = 1$ gives, after simplification,

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$$\left. \frac{\partial F_{F,S}}{\partial w} \right|_{w=1} = \frac{2z^2}{(1 - 3z)^2}.$$

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1559 Hence

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1561

$$E(n) \cdot 3^n = [z^n] \frac{2z^2}{(1 - 3z)^2} = 2(n - 1)3^{n-2}, \quad n \geq 2,$$

1562 giving $E(n) = 2(n - 1)/9$ exactly, with no exponential correction term. This matches

1563

1564 the linearity-of-expectation derivation given below.

Verification. For $n = 3$, brute-force enumeration over all $3^3 = 27$ amyloids in $\{C, Y, S\}^3$ gives $\mathcal{A}(3, 0) = 17$, $\mathcal{A}(3, 1) = 8$, and $\mathcal{A}(3, 2) = 2$, with $E(3) = (0 \cdot 17 + 1 \cdot 8 + 2 \cdot 2)/27 = 12/27 = 4/9$. This agrees with both the linearity-of-expectation formula $E(3) = 2(3 - 1)/9 = 4/9$ and the generating-function expansion: $[z^3]F_{F,S}(z, w) = 17 + 8w + 2w^2$ by direct computation of the recurrence $f_n = (2 + w)f_{n-1} + (1 - w)f_{n-2}$. For $n = 4$, both routes give $\mathcal{A}(4, k) = 41, 28, 10, 2$ for $k = 0, 1, 2, 3$, with $E(4) = (28 + 20 + 6)/81 = 54/81 = 2/3 = 2 \cdot 3/9$ as expected. (See `verify_section5_gf.py`.)

The bivariate generating function above encodes the full probability distribution via $P(n, k) = \mathcal{A}(n, k)/3^n$. However, for computing the expected value, a much simpler route is available. As in Section 4, the total yellow fluorescence of an amyloid equals the number of adjacent $\{C, Y\}$ pairs. Since FRET interactions have no blocking constraint (unlike split-YFP fusion), the linearity of expectation argument from Section 4 applies directly. With each protein independently C, Y, or S with probability $1/3$:

$$E(n) = (n - 1) \cdot P(\{C, Y\} \text{ pair}) = (n - 1) \cdot 2 \cdot \frac{1}{3} \cdot \frac{1}{3} = \frac{2(n - 1)}{9}$$

Therefore

$$R_{F,S}(n) := E(n)/n = \frac{2(n - 1)}{9n} \longrightarrow \frac{2}{9} \text{ as } n \rightarrow \infty$$

The first 400 values of $R_{F,S}(n)$ appear in Figure 4.

6 Variance, null-model fluctuations, and finite-radius FRET

6.1 Finite- n variances

The preceding sections emphasize exact expectations, but the same probability distributions also give exact finite- n fluctuations. Let K_n denote the fluorescence count

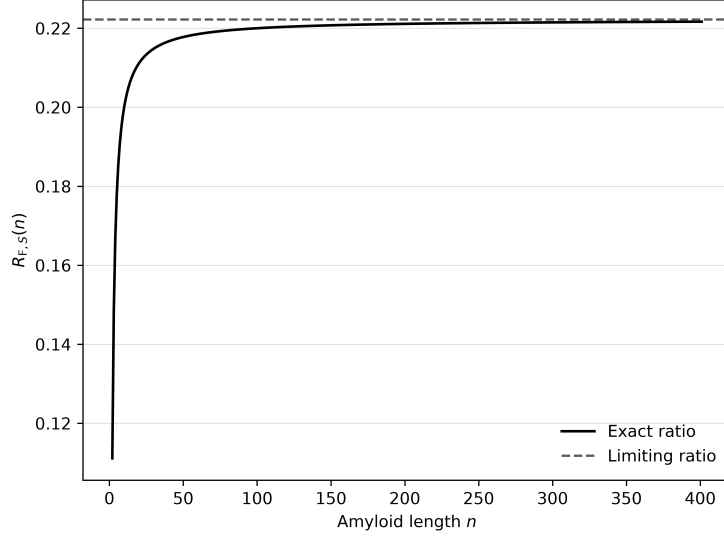


Fig. 4 $R_{F,S}(n)$ for FRET YFP/CFP amyloids with Sup35p. The dashed horizontal line is $2/9$ and the solid curve is the exact $R_{F,S}(n) = 2(n-1)/(9n)$. Convergence is algebraic ($O(1/n)$), as in Figure 3.

in the relevant model. If

$$F(z, w) = \sum_{n \geq 0} \sum_{k \geq 0} \mathcal{A}(n, k) w^k z^n$$

is the corresponding bivariate generating function and T_n is the total number of amyloids of length n , then

$$E[K_n] = \frac{[z^n] F_w(z, 1)}{T_n}, \quad E[K_n(K_n - 1)] = \frac{[z^n] F_{ww}(z, 1)}{T_n},$$

and hence

$$\text{Var}(K_n) = E[K_n(K_n - 1)] + E[K_n] - E[K_n]^2.$$

The variance formulas below are stated for $n \geq 2$; for $n = 1$ one has $K_1 = 0$ and variance 0 in all four models. For reference, the model sizes and bivariate generating

functions used in this section are:

Model	T_n	$F(z, w)$
split-YFP	2^n	$\frac{1+z}{1-z-2wz^2}$
split-YFP+Sup35p	3^n	$\frac{1+z}{1-2z-(1+2w)z^2}$
FRET	2^n	$\frac{1+z(1-w)}{1-z(1+w)}$
FRET+Sup35p	3^n	$\frac{1+(1-w)z}{1-(2+w)z-(1-w)z^2}$

For the exclusively split-YFP model, the bivariate generating function is

$$F_{\text{SY}}(z, w) = \frac{1+z}{1-z-2wz^2},$$

obtained from the same two-state decomposition as in Section 2. In transfer-matrix form, with p denoting an unfused A/B endpoint and q denoting a fused endpoint, the weighted transition matrix is

$$M_{\text{SY}}(w) = \begin{pmatrix} 1 & w \\ 2 & 0 \end{pmatrix},$$

where the rows and columns are ordered as (p, q) and w marks a fusion. Including the empty amyloid and the initial vector $(2, 0)$ gives

$$F_{\text{SY}}(z, w) = 1 + z(2, 0) (I - zM_{\text{SY}}(w))^{-1} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \frac{1+z}{1-z-2wz^2}.$$

Its second derivative at $w = 1$ is

$$\left. \frac{\partial^2 F_{\text{SY}}}{\partial w^2} \right|_{w=1} = \frac{8z^4}{(1-2z)^3(1+z)^2}.$$

1703 Using this together with the expectation from Section 2 gives, for $n \geq 2$,

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$$\text{Var}_{\text{SY}}(K_n) = \frac{6n+2}{81} + \frac{(12n+2)(-1)^n}{81 \cdot 2^n} - \frac{4}{81 \cdot 4^n}.$$

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1710 For split-YFP with Sup35p, differentiating (12) twice gives

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$$\left. \frac{\partial^2 F}{\partial w^2} \right|_{w=1} = \frac{8z^4}{(1-3z)^3(1+z)^2},$$

1715

1716 and therefore

1717

1718

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$$\text{Var}_{\text{SY},S}(K_n) = \frac{56n-27}{576} + \frac{(4n+3)(-1)^n}{48 \cdot 3^n} - \frac{1}{64 \cdot 9^n}, \quad n \geq 2.$$

1720

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1723 For FRET without Sup35p, Section 4 identifies K_n as a Binomial($n-1, 1/2$) random

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1725 variable, so

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$$\text{Var}_{\text{F}}(K_n) = \frac{n-1}{4}.$$

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1735 which yields

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$$\text{Var}_{\text{F},S}(K_n) = \frac{18n-22}{81}, \quad n \geq 2.$$

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1739 As a small check, at $n = 2$ the split-YFP-with-Sup35p and FRET-with-Sup35p dis-

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tributions have $P(K_2 = 0) = 7/9$ and $P(K_2 = 1) = 2/9$, giving variance $14/81$, exactly as the formulas above give. In the no-Sup35p split-YFP case, $P(K_2 = 0) = P(K_2 = 1) = 1/2$, while at $n = 3$ one has $P(K_3 = 0) = 2/8$ and $P(K_3 = 1) = 6/8$, giving variances $1/4$ and $3/16$, again matching the formula. The accompanying script

`verify_variance.py` brute-force checks all four variance formulas against direct enumeration of $P(n, k)$ for $2 \leq n \leq 7$.

These formulas give exact intrinsic fluctuation scales under the i.i.d. null model; they are not, by themselves, experimental measurement error bars. Since the split-YFP ratios are $2K_n/n$ and the FRET ratios are K_n/n ,

$$\text{Var}(R_{\text{SY}}(n)) = \frac{4}{n^2} \text{Var}_{\text{SY}}(K_n), \quad \text{Var}(R_{\text{SY},S}(n)) = \frac{4}{n^2} \text{Var}_{\text{SY},S}(K_n)$$

$$\text{Var}(R_{\text{F}}(n)) = \frac{1}{n^2} \text{Var}_{\text{F}}(K_n), \quad \text{Var}(R_{\text{F},S}(n)) = \frac{1}{n^2} \text{Var}_{\text{F},S}(K_n)$$

Thus, for large n , the standard deviations of the ratio statistics scale as $O(n^{-1/2})$.

For example,

$$\begin{aligned} \text{Var}(R_{\text{SY}}(n)) &\sim \frac{8}{27n}, & \text{Var}(R_{\text{SY},S}(n)) &\sim \frac{7}{18n}, \\ \text{Var}(R_{\text{F}}(n)) &\sim \frac{1}{4n}, & \text{Var}(R_{\text{F},S}(n)) &\sim \frac{2}{9n}. \end{aligned}$$

This is the usual central-limit scale for additive statistics of finite-state chains and transfer matrices. More formal Gaussian limit laws could be obtained from standard transfer-matrix or quasi-power arguments, but the exact finite- n distributions and variances above are the quantities used here. For short amyloids, the exact distributions $P(n, k)$ should be used directly. For experimental ensembles of many fibrils, the relevant uncertainty also depends on the sampling design, the number of fibrils measured, and instrumental noise.

6.2 Finite-radius FRET expectation

The nearest-neighbor FRET assumption can also be relaxed at the level of expected value. Suppose a YFP can receive FRET from CFPs within distance r along the amyloid, with additive unit contribution from each CFP–YFP pair in this window.

1795 Let p_C and p_Y be the probabilities of CFP and YFP, and set $m = \min(r, n - 1)$.
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1797 There are $n - d$ unordered pairs at distance d , each contributing in expectation $2p_C p_Y$.
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1799 Linearity of expectation requires no independence among these pair indicators, so the
1800 expectation is exact even though the indicators are correlated. Hence

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1802

$$1803 \quad E_r(n) = 2p_C p_Y \sum_{d=1}^m (n - d) = 2p_C p_Y \left(mn - \frac{m(m+1)}{2} \right).$$

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1807 For fixed r and $n \rightarrow \infty$, this gives

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$$1810 \quad \frac{E_r(n)}{n} \longrightarrow 2rp_C p_Y.$$

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1813 In the equal-concentration FRET+Sup35p case ($p_C = p_Y = 1/3$), the nearest-
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1815 neighbor model $r = 1$ recovers $E_1(n) = 2(n - 1)/9$, while $r = 2$ gives

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$$1818 \quad E_2(n) = \frac{2}{9}((n - 1) + (n - 2)) = \frac{4n - 6}{9}, \quad n \geq 3,$$

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1822 and $E_2(n)/n \rightarrow 4/9$. A crude hard-cutoff surrogate motivated by $R_0/0.47 \text{ nm} \approx 10$
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1824 would scale this unit-window baseline linearly in r , but that number is a property of
1825 the idealized indicator sum, not a FRET-efficiency-weighted prediction. The point is
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1827 only that the nearest-neighbor value $2/9$ is an $r = 1$ mathematical baseline rather
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1829 than a quantitative experimental prediction.

1830 The unit-weight window is itself an idealization: a more physical model could
1831
1832 weight a CFP–YFP pair at distance d by a distance-dependent FRET efficiency η_d
1833
1834 rather than by a hard cutoff (for example, an efficiency kernel depending on distance
1835 and orientation). The expected-value computation would still go through, giving

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$$1838 \quad \frac{E(n)}{n} \longrightarrow 2p_C p_Y \sum_{d \geq 1} \eta_d$$

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1840

whenever the efficiency weights are summable, but the closed form would depend on the chosen geometry. Computing the full distribution for $r > 1$ is substantially harder: adjacent indicators are correlated over a larger window, and a transfer matrix would need to track the last r protein identities rather than only the rightmost state. For $n = 2$, the value $m = \min(r, n - 1)$ is still 1 even when $r = 2$, so the radius-two case collapses to the nearest-neighbor formula until $n \geq 3$.

7 Generalization to unequal concentrations

The results of Sections 2–5 all depend on the equal-concentration assumption. The Markov chain arguments generalize readily to unequal concentrations, which we outline here. This section records limiting per-step fluorescence ratios rather than full finite- n distributions. Such distributions could be obtained by the same finite-state transfer-matrix method, with probabilities replacing the equal-count weights, but the resulting formulas are less compact and are left as a natural extension for experimental calibration.

7.1 Split-YFP with unequal concentrations

Suppose protein A appears with probability α and protein B with probability $1 - \alpha$. A two-state chain on $\{\text{unfused}, \text{fused}\}$ is insufficient here, because the probability that the next protein triggers a fusion conditional on the current rightmost being unfused depends on whether the rightmost is an A or a B: from rightmost-A, fusion requires next = B (probability $1 - \alpha$), while from rightmost-B it requires next = A (probability α). For $\alpha \neq 1/2$ these are not equal, so we cannot lump A-unfused and B-unfused into a single state without loss. We instead use the three-state chain $\{p_A, p_B, q\}$ from Section 7.2, specialized to the no-Sup35p case $\beta = 1 - \alpha$ (so $1 - \alpha - \beta = 0$). The transitions are:

- From p_A : next is A (prob α , stay p_A); next is B (prob $1 - \alpha$, fuse, go q).

1887 • From p_B : next is B (prob $1-\alpha$, stay p_B); next is A (prob α , fuse, go q).
 1888
 1889 • From q : next is A (prob α , go p_A); next is B (prob $1-\alpha$, go p_B).
 1890
 1891 The balance equations $\pi_A = \alpha\pi_A + \alpha\pi_q$ and $\pi_B = (1-\alpha)\pi_B + (1-\alpha)\pi_q$, together with
 1892 $\pi_A + \pi_B + \pi_q = 1$, give
 1893
 1894
 1895
$$\pi_A = \frac{\alpha^2}{1 - \alpha(1 - \alpha)}, \quad \pi_B = \frac{(1 - \alpha)^2}{1 - \alpha(1 - \alpha)}, \quad \pi_q = \frac{\alpha(1 - \alpha)}{1 - \alpha(1 - \alpha)}.$$

 1896
 1897
 1898
 1899 The expected fluorescence per step is $e = (1-\alpha)\pi_A + \alpha\pi_B = \alpha(1-\alpha)/(1-\alpha(1-\alpha))$, so
 1900
 1901
 1902
$$\lim_{n \rightarrow \infty} R_{SY}(n) = 2e = \frac{2\alpha(1 - \alpha)}{1 - \alpha(1 - \alpha)}.$$

 1903
 1904
 1905
 1906 Equivalently, this is the specialization $\beta = 1 - \alpha$ of the Section 7.2 formula $2\alpha\beta(2 -$
 1907 $\alpha - \beta)/(1 - \alpha\beta)$. At $\alpha = 1/2$, $R_{SY} \rightarrow (1/2)/(3/4) = 2/3$, recovering Section 2. The
 1908
 1909 maximum occurs at equal concentrations: differentiating gives
 1910
 1911
 1912
$$\frac{d}{d\alpha} \left(\frac{2\alpha(1 - \alpha)}{1 - \alpha(1 - \alpha)} \right) = \frac{2(1 - 2\alpha)}{(1 - \alpha(1 - \alpha))^2},$$

 1913
 1914
 1915
 1916 so the only interior critical point is $\alpha = 1/2$, and the endpoints have zero fluorescence.
 1917 Numerically, at $\alpha = 1/4$ the limit is $2 \cdot \frac{3}{16} / \frac{13}{16} = 6/13 \approx 0.462$, in agreement with
 1918
 1919 Monte Carlo simulation (the default script run with $n = 5000$ and 400 trials gives
 1920
 1921 approximately 0.462; see `split_yfp_unequal_simulation.py`). A naive two-state
 1922 chain would wrongly lump p_A and p_B into a single unfused state and assign fusion
 1923 probability $d = 2\alpha(1 - \alpha)$ from that lumped state, giving the ratio $2d/(1 + d) =$
 1924 $4\alpha(1 - \alpha)/(1 + 2\alpha(1 - \alpha))$; at $\alpha = 1/4$ this is $6/11 \approx 0.545$. The two-state lumping is
 1925
 1926 harmless at $\alpha = 1/2$, where A/B symmetry gives equal transition probabilities from
 1927
 1928 p_A and p_B and hence a genuinely lumpable chain, but it fails the simulation check at
 1929
 1930 every $\alpha \neq 1/2$.
 1931
 1932

The convergence rate of $R_{SY}(n)$ to its limit depends on α . Near $\alpha = 0$ or $\alpha = 1$, the chain changes state rarely and finite- n values can differ noticeably from the limiting ratio. For experimental fibril lengths, the finite- n value obtained by iterating the three-state chain or by simulation should be used rather than the stationary limit alone.

7.2 Split-YFP with Sup35p at unequal concentrations

Now suppose proteins A, B, S appear with probabilities α , β , and $1-\alpha-\beta$ respectively. We define three Markov states: p_A (rightmost is A, unfused), p_B (rightmost is B, unfused), and q (rightmost is S, or fused). The transitions are:

- From p_A : next is A (prob α , stay p_A), B (prob β , fuse, go q), or S (prob $1-\alpha-\beta$, go q).
- From p_B : next is B (prob β , stay p_B), A (prob α , fuse, go q), or S (prob $1-\alpha-\beta$, go q).
- From q : next is A (prob α , go p_A), B (prob β , go p_B), or S (prob $1-\alpha-\beta$, stay q).

Solving the balance equations $\pi_A = \alpha\pi_A + \alpha\pi_q$, $\pi_B = \beta\pi_B + \beta\pi_q$, and $\pi_A + \pi_B + \pi_q = 1$ gives $\pi_A = \alpha(1-\beta)/(1-\alpha\beta)$, $\pi_B = \beta(1-\alpha)/(1-\alpha\beta)$, and $\pi_q = (1-\alpha)(1-\beta)/(1-\alpha\beta)$. The expected fluorescence per step is $e = \beta\pi_A + \alpha\pi_B = \alpha\beta(2-\alpha-\beta)/(1-\alpha\beta)$, giving

$$\lim_{n \rightarrow \infty} R_{SY,S}(n) = 2e = \frac{2\alpha\beta(2-\alpha-\beta)}{1-\alpha\beta}$$

which reduces to $1/3$ when $\alpha = \beta = 1/3$ (and to $2/3$ when $\alpha = \beta = 1/2$, recovering the no-Sup35p case). The same simulation script also checks this unequal-Sup35p limit for representative (α, β) pairs.

7.3 FRET with unequal concentrations

In the two-species FRET setting, $P(C) = p_C$ and $P(Y) = p_Y = 1 - p_C$, so linearity of expectation gives $R_F(n) \rightarrow 2p_C p_Y$ directly. In the three-species FRET+Sup35p setting, $P(C) = p_C$, $P(Y) = p_Y$, and $P(S) = p_S$ with $p_C + p_Y + p_S = 1$; the same adjacent-pair calculation gives $R_{F,S}(n) \rightarrow 2p_C p_Y$. These generalizations would allow the model to be calibrated to experimental concentration ratios.

8 Conclusion

We have derived exact probability distributions, closed-form expected values, and finite- n variances for fluorescence in four combinatorial models of amyloid fibrils. The exact expectation formulas in Table 5 are for $n \geq 1$, with $E(1) = 0$ in all four models; the $n = 0$ terms used in generating functions are formal conventions only. The principal expectation results are summarized in each model's native units.

Table 5 Exact expectations and limiting native ratios for the four models

System	Native ratio	Exact $E(n)$	Limit
Split-YFP (Sec. 2)	$R_{SY} = 2E/n$	$(3n - 2)/9 + 2(-1)^n/(9 \cdot 2^n)$	$2/3$
Split-YFP + Sup35p (Sec. 3)	$R_{SY,S} = 2E/n$	$(4n - 3)/24 - (-1)^{n-1}/(24 \cdot 3^{n-1})$	$1/3$
FRET YFP/CFP (Sec. 4)	$R_F = E/n$	$(n - 1)/2$	$1/2$
FRET + Sup35p (Sec. 5)	$R_{F,S} = E/n$	$2(n - 1)/9$	$2/9$

Here R_{SY} and $R_{SY,S}$ are fractions of proteins participating in split-YFP fluorescence (two proteins per fused pair), while R_F and $R_{F,S}$ are total FRET fluorescence units per protein. The table places the four exact baselines side by side for reference, but the split-YFP and FRET ratios measure different physical quantities and are not directly comparable.

A key structural distinction separates the two pairs of systems. In the split-YFP settings (Sections 2 and 3), the adjacency blocking constraint—a fused pair cannot participate in further fusion—introduces nontrivial combinatorial structure. This necessitates generating function techniques to derive the full distributions $P(n, k)$ and Markov chain arguments to compute the expected values in closed form. The agreement between these two independent approaches provides mutual verification of both the sum identities and the Markov chain analysis.

In the FRET settings (Sections 4 and 5), the absence of a blocking constraint means that the total fluorescence is a sum of adjacent-pair contributions: one interaction does not consume a protein and prevent another interaction. Thus a direct linearity-of-expectation argument gives the exact expectation. In the two-letter FRET model these adjacent-pair indicators are independent by the bijection in Section 4, while in the Sup35p model they are correlated, as reflected in the variance formula.

The limiting ratios admit clean interpretations under the i.i.d. null model. For split-YFP amyloids under random assembly, $R_{SY}(n) \rightarrow 2/3$ means that in a long amyloid composed of equal concentrations of the two halves, approximately two-thirds of the proteins participate in a fluorescent fusion—a high baseline fluorescence yield arising from the alternating tendency of random binary sequences. Introducing Sup35p at equal concentration gives $R_{SY,S}(n) \rightarrow 1/3$: the inert Sup35p proteins act as spacers that interrupt potential fusion pairs. For FRET, the ratio $R_F(n) \rightarrow 1/2$ reflects the null-model probability that a random adjacent pair is a $\{C, Y\}$ pair, while $R_{F,S}(n) \rightarrow 2/9$ reflects the reduced probability $2 \cdot (1/3)^2$ of such a pair when three protein types compete for each position. The variance formulas in Section 6 give finite- n null-model fluctuation scales, so experimentally measured fluorescence ratios can be compared against both the expected value and the size of random fluctuations under the i.i.d. model.

2071 Section 6 also records the expected-value extension of the FRET model to an
2072 interaction radius r . This partially addresses the nearest-neighbor simplification while
2073 preserving a closed-form baseline: the expectation remains a direct sum over all CFP–
2074 YFP pairs within distance r . A hard cutoff with r on the order of ten amyloid spacings
2075 would scale the unit-window expectation far above the nearest-neighbor value, under-
2076 scoring that the $r = 1$ model is a mathematically transparent baseline rather than a
2077 quantitative FRET prediction. However, the full distribution for $r > 1$ would require
2078 generating functions or transfer matrices for strings with longer-range dependencies,
2079 a substantially richer combinatorial problem.

2086 Similarly, Section 7 gives limiting ratios for unequal concentrations but not the
2087 corresponding finite- n distributions or variances. Those distributions are accessible in
2088 principle through larger or weighted transfer matrices, and would be the next step for
2089 fitting the null model to measured concentration ratios.

2092 Our analysis assumes that proteins are independently distributed in the amyloid
2093 sequence. This assumption neglects known cooperative and templated growth effects
2094 in amyloid assembly (Chiti and Dobson 2006) and should be interpreted as a com-
2095 binatorial null model: a baseline against which experimental deviations—reflecting
2096 assembly biases, kinetic trapping, or cooperative interactions—could be quantified.
2097 The combinatorial framework developed here, particularly the generating function
2098 approach to constrained fluorescence patterns via regular expressions, could in princi-
2099 ple be adapted to non-uniform protein distributions if experimental data on assembly
2100 biases become available.

2107 All simulation and verification scripts referenced in this paper are available
2108 at <https://github.com/danielchen0/amyloids>. These include the split-YFP, Sup35p,
2109 FRET, unequal-concentration, variance, and generating function verification scripts,
2110 together with the associated plotting scripts.

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at https://github.com/danielchen0/amyloids .	2135
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